

Suppose now that (ii) holds. We may assume that  $x(s) \leq x(s')$ . Lemma 12.5.6 now gives  $x(c') + x(d') < x(s) + x(s') \leq 2x(s')$ , so  $x(c') < x(s')$  or  $x(d') < x(s')$ . In either case  $s'$  is not  $\mathcal{S}$ -friendly, by  $c', d' \in S'$  and our choice of  $S'$  for  $s'$ .  $\square$

The following general theorem about entanglements in graphs implies Theorem 12.5.1. Indeed, take as the  $\mathcal{S}$  in Theorem 12.5.8 the set of all sets  $D(\tau, \tau')$  of efficient distinguishers of distinguishable tangles in  $G$ ; recall that, by Lemma 12.5.4, these sets  $D(\tau, \tau')$  are entanglements.

**Theorem 12.5.8.** (Carmesin & Kurkofka, 2024)

*Let  $\mathcal{S}$  be any set of entanglements in a graph  $G$ . Then the  $\mathcal{S}$ -friendly separations of  $G$  are nested.*

*Proof.* It suffices to show that  $\mathcal{S}$  satisfies the premise of Lemma 12.5.7. So consider  $S, S' \in \mathcal{S}$  with crossing separations  $s \in S$  and  $s' \in S'$  as in the lemma, with  $|s| \leq |s'|$  say. Let us call a corner of  $s$  and  $s'$  *relevant* if it has order at most  $|s'|$ . Let us show the following:

*At least three corners of  $s$  and  $s'$  are relevant.* (1)

Suppose at most two corners of  $s$  and  $s'$  are relevant. By submodularity (†) and  $|s| \leq |s'|$ , at least one of any two opposite corners must be relevant, so there are adjacent relevant corners. As their opposite two corners are irrelevant and thus have order  $> |s'|$ , our adjacent relevant corners have order  $< |s|$  by submodularity. If they lie on the same side of  $s$ , then this contradicts our assumption that  $S$  is an entanglement. If they lie on the same side of  $s'$ , it contradicts our assumption that  $S'$  is an entanglement. This completes the proof of (1).

*We may assume that all four corners of  $s$  and  $s'$  are relevant.* (2)

To prove (2), suppose only three corners are relevant. Two of these,  $c'$  and  $d'$  say, lie on the same side of  $s'$ . One of them,  $c'$  say, then lies in  $S'$  and has order exactly  $|s'|$ , by (E) for  $S'$ . By submodularity, the corner  $c$  opposite  $c'$  has order at most  $|s|$ ; in particular,  $c$  is relevant too (Fig. 12.5.3). By assumption, the remaining corner  $d$  is irrelevant and hence has order  $> |s'|$ . By submodularity, therefore, its opposite corner  $d'$  has order  $< |s|$ . Condition (E) applied to  $S$  and  $s$  now yields that  $|c| = |s|$  and  $c \in S$ . We thus have (i) of Lemma 12.5.7, as desired. This completes the proof of (2).

By (E) for  $S'$ , the separation  $s'$  has corners on both sides that have order  $|s'|$  and lie in  $S'$ . Assume first that these are adjacent, say  $c'$  and  $d$ . Let  $c$  be the corner opposite  $c'$ , and  $d'$  that opposite  $d$ . By submodularity,  $c$  and  $d'$  have order at most  $|s|$ . As they lie on the same side of  $s$ , one of them lies in  $S$ , by (E) for  $S$ . Since its opposite corner lies in  $S'$ , we have (i) of Lemma 12.5.7 as desired.

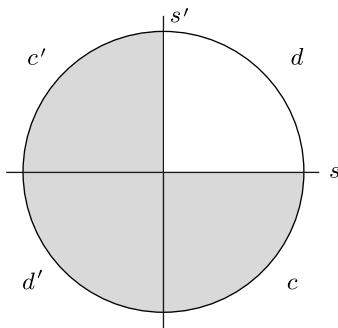


Fig. 12.5.3. Corners of  $s$  and  $s'$  in the proof of Theorem 12.5.8.  
Relevant corners in the proof of (2) are shaded.

Suppose now that the two corners on the different sides of  $s'$  that have order  $|s'|$  and lie in  $S'$  are opposite corners, say  $c'$  and  $c$  in Figure 12.5.3. By submodularity, this means that our assumption of  $|s| \leq |s'|$  must hold with equality:  $|s| = |s'|$ . Applying (E) to the side of  $s$  on which  $d$  lies, we find that  $d$  or  $c'$  lies in  $S$ . If  $c'$  does, then  $c' \in S$  and  $c \in S'$  satisfy (i) of Lemma 12.5.7. So we assume that  $d \in S$ .

Applying (E) to the other side of  $s$ , we find that  $c$  or  $d'$  lies in  $S$ . If  $c$  does, then again  $c$  and  $c'$  satisfy (i) of Lemma 12.5.7. So we assume that  $d' \in S$ . We now have (ii) of Lemma 12.5.7, with  $d, d' \in S$  and  $c, c' \in S'$ . This completes our verification of the premise of Lemma 12.5.7.  $\square$

Theorem 12.5.8, like Theorem 12.5.1, tells us how all the tangles in a graph  $G$  are distinguished by a certain set  $N$  of nested separations. These are induced by a tree-decomposition of  $G$  (Exercises 16, 51). More generally, for every  $k \in \mathbb{N}$  the set  $N_k$  of separations of order  $< k$  in  $N$  is induced by a tree-decomposition  $(T_k, \mathcal{V}_k)$  of  $G$ . Then every  $k$ -tangle  $\tau$  in  $G$  orients all the separations in  $N_k$  towards  $V_t$  for some  $t \in T_k$  depending on  $\tau$ , where  $\mathcal{V}_k = (V_t)_{t \in T_k}$  as usual. We may think of the tangle  $\tau$  as ‘living in’ this node  $t$  of  $T_k$ , or in the bag  $V_t$  it specifies. For  $i < j$ , the tree  $T_i$  is obtained from  $T_j$  by contracting the edges that correspond, as in Lemma 12.3.1, to separations in  $N_j \setminus N_i$ , while the bags of  $\mathcal{V}_i$  are unions of bags of  $\mathcal{V}_j$  indexed by the  $t \in T_j$  that lie in a common branch set of this contraction minor  $T_i \preceq T_j$ .

The tree-structure which a nested set of separations imposes on  $G$  can also be made visible in a more direct way, without a detour via tree-decompositions, as follows.

The set  $\vec{E}(T)$  of the oriented edges of a tree  $T$  is partially ordered by letting  $\vec{e} \geq \vec{f}$  whenever  $\vec{e} = (e, x, y)$  and  $\vec{f} = (f, u, v)$  are such that the unique  $\{x, y\} - \{u, v\}$  path in  $T$  starts in  $y$  and ends in  $u$ . Note that every two edges of  $T$  are ‘nested’ in our earlier sense that they have comparable orientations. Call a subset  $\tau$  of  $\vec{E}(T)$  *consistent* if it contains no  $\vec{e}$  such

$$\vec{e} \geq \vec{f}$$

consistent

that  $\vec{e} \geq \vec{f} \in \tau$  for some  $f$ . The consistent orientations of  $E(T)$  are precisely those towards some node  $t$  of  $T$  (Ex. 50). The set of minimal elements of such an orientation, then, is the set

$$\sigma_t := \{ (e, s, t) \in \vec{E}(T) \mid e = st \in T \}$$

of all the incoming edges at  $t$ .

Given a set  $S$  of separations of  $G$ , an *S-tree* is a pair  $(T, \alpha)$  such that  $T$  is a tree and  $\alpha: \vec{E}(T) \rightarrow \vec{S}$  respects the orderings on these sets and commutes with inversion:  $\alpha(\vec{e}) \geq \alpha(\vec{f})$  if  $\vec{e} \geq \vec{f}$  (Fig. 12.5.4), and  $\alpha(\vec{e}) = (B, A)$  whenever  $\alpha(\vec{e}) = (A, B)$ .<sup>9</sup> We then say that  $(T, \alpha)$  is an *S-tree over* a set  $\mathcal{F}$  of subsets of  $\vec{S}$  if  $\alpha(\sigma_t) \in \mathcal{F}$  for every node  $t$  of  $T$ . Note that we do not require  $\alpha$  to be injective, not even on the sets  $\sigma_t$ .

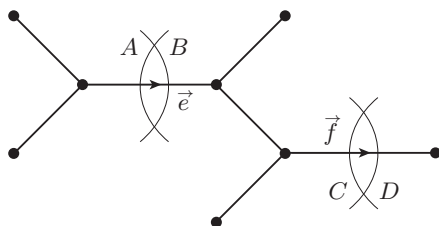


Fig. 12.5.4. An *S-tree* with  $\alpha(\vec{e}) = (A, B) \geq (C, D) = \alpha(\vec{f})$

As every two edges of a tree have comparable orientations, the image of  $\vec{E}(T)$  under  $\alpha$  for an *S-tree*  $(T, \alpha)$  defines a nested subset of  $S$ . Conversely, every nested set  $N$  of separations of  $G$  underlies the image of  $\alpha$  for a suitable *S-tree*  $(T, \alpha)$ , with  $S = N$  say. For example, a set  $N$  of efficient tangle-distinguishers as in Theorem 12.5.1 can be displayed by an *N-tree*  $(T, \alpha)$  in this way. Its minors  $T_1 \preceq T_2 \preceq \dots$  defined earlier, then, display the sets  $N_k$  of efficient tangle-distinguishers of order  $< k$  in  $N$  for  $k = 1, 2, \dots$ : those that distinguish, among others, the  $k$ -tangles in  $G$ .

The second main theorem about tangles, the *tangle-tree duality* theorem, employs  $S_k$ -trees to display the tree-structure of graphs that have no  $k$ -tangles for some desired  $k$ . Intuitively, the bags of the corresponding tree-decompositions will be ‘too small’ to be home to a tangle. To express this directly in terms of *S-trees*, we need another definition.

Recall that the *k-tangles* in  $G$  are the orientations of  $S_k$  that avoid  $\mathcal{T}$ . A set  $\sigma$  of oriented separations is a *star* if  $(A, B) \geq (D, C)$  for all distinct  $(A, B), (C, D) \in \sigma$  (Fig. 12.5.5). Let  $\mathcal{T}^* := \{ \sigma \in \mathcal{T} \mid \sigma \text{ is a star} \}$ .

As we shall see in a moment,  $S_k$ -trees over  $\mathcal{T}^*$  are natural certificates for the non-existence of  $k$ -tangles. The *tangle-tree duality* theorem guarantees that these certificates exist whenever a graph has no  $k$ -tangle:

<sup>9</sup> A tree-decomposition  $(T, \mathcal{V})$ , for example, makes  $T$  into an *S-tree* for the set  $S$  of separations it induces.

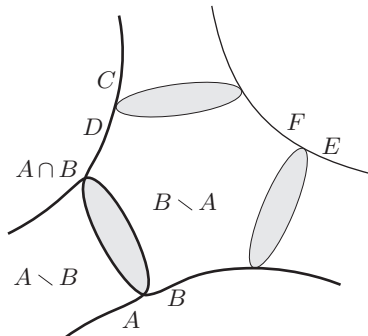


Fig. 12.5.5. The separations  $(A, B), (C, D), (E, F)$  form a star

**Theorem 12.5.9.** (Robertson & Seymour 1991)

tangle-tree  
duality  
theorem

The following assertions are equivalent for all graphs  $G$  and integers  $k > 0$ :

- (i)  $G$  has no tangle of order  $k$ ;
- (ii)  $G$  has an  $S_k$ -tree over  $\mathcal{T}^*$ .

**Lemma 12.5.10.** (Uncrossing Lemma)

Every consistent orientation of  $S_k$  that has a subset  $\sigma \in \mathcal{T}$  also has a subset in  $\mathcal{T}^*$ .

*Proof.* Let  $\sigma = \{(A_i, B_i) \mid i = 1, 2, 3\} \in \mathcal{T}$  be contained in a consistent orientation  $O$  of  $S_k$ . We show that, unless  $\sigma$  is a star, we can replace one of its elements by a strictly greater separation in  $O$  while keeping it in  $\mathcal{T}$ . In finitely many steps this will turn  $\sigma$  into a star: a subset of  $O$  in  $\mathcal{T}^*$ .

If  $\sigma$  is not a star we may assume that  $(A_1, B_1) \not\geq (B_2, A_2)$ . By  $(\dagger)$  and  $\sigma \subseteq \vec{S}_k$  we may further assume that  $(C, D) := (A_1 \cap B_2, B_1 \cup A_2) \in \vec{S}_k$ ; the other case, that  $(B_1 \cap A_2, A_1 \cup B_2) \in \vec{S}_k$ , is analogous. Since  $O$  is consistent and  $(C, D) \geq (A_1, B_1) \in O$ , we cannot have  $(D, C) \in O$ . Thus,  $(C, D) \in O$ . But  $(C, D) > (A_1, B_1)$ , since either  $A_1 \not\subseteq B_2$  or  $B_1 \not\subseteq A_2$  by assumption. Replacing  $(A_1, B_1)$  in  $\sigma$  with  $(C, D)$  to obtain  $\sigma'$ , say, gives the desired reduction: since any vertex or edge of  $G[A_1]$  that does not lie in  $G[C]$  lies in  $G[A_2]$ , and  $(A_2, B_2) \in \sigma'$ , we have  $\sigma' \in \mathcal{T}$ .  $\square$

**Proof of Theorem 12.5.9.** (ii) $\rightarrow$ (i) Suppose  $G$  has an  $S_k$ -tree  $(T, \alpha)$  over  $\mathcal{T}^*$ . Then any  $k$ -tangle  $\tau$  in  $G$  defines via  $\alpha$  an orientation of the edges of  $T$ . Let  $t \in T$  be the last node of any maximal path in  $T$  whose edges are all oriented forward. Then all the edges at  $t$  are oriented towards  $t$ , and  $\alpha$  maps these oriented edges to a star in  $\mathcal{T}$ . In particular,  $\tau$  has a subset in  $\mathcal{T}$  and thus cannot be a tangle.

(i) $\rightarrow$ (ii) To make this implication susceptible to an induction proof, our strategy is to relax both (i) and (ii) in some carefully designed way that makes it easier to prove the implication at the induction start. The

relaxation, however, gets a little less in each inductive step, so that at the end it has evaporated and we have our desired implication of (i)→(ii). More precisely, we shall prove that certain sets  $\mathcal{T}^+$  satisfy one of

- (1)  $S_k$  has a consistent orientation that avoids  $\mathcal{T}^* \cup \mathcal{T}^+$ ;
- (2)  $G$  has an  $S_k$ -tree over  $\mathcal{T}^* \cup \mathcal{T}^+$ .

This yields (i)→(ii) when  $\mathcal{T}^+ = \emptyset$ . Indeed, if  $G$  has no  $k$ -tangle then, by Lemma 12.5.10, (1) cannot hold. So (2) must hold, but (2) is just (ii) when  $\mathcal{T}^+ = \emptyset$ .

The sets  $\mathcal{T}^+$  for which we prove this will depend on certain sets  $O$  of oriented separations, which we shall vary in the induction. Given  $O$ , let  $\mathcal{T}^+ := \{(B, A)\} : (A, B) \in O\} \setminus \mathcal{T}^*$ . Thus,  $S_k$ -trees over  $\mathcal{T}^* \cup \mathcal{T}^+$  are given more freedom compared with  $S_k$ -trees over  $\mathcal{T}^*$ : singleton stars associated with their leaves need not lie in  $\mathcal{T}^*$ , as long as they lie in  $\mathcal{T}^+$ . This happens when the inverse of their element lies in  $O$  but is not of the form  $(A, V)$ . In the latter case,  $\{(V, A)\}$  lies in  $\mathcal{T}^*$ , and hence not in  $\mathcal{T}^+$ .

The sets  $O$  we consider in our induction will be all the sets  $O \subseteq \overrightarrow{S}_k$  that have the following two properties common to all  $k$ -tangles. First, they will contain all the separations  $(A, V)$  with  $|A| < k$ .

Second, they will be *closed upwards* in  $\overrightarrow{S}_k$ : they contain any  $(A, B) \in \overrightarrow{S}_k$  such that  $(A, B) \geq (C, D)$  for some  $(C, D) \in O$ . In particular, our induction covers the smallest such set,  $O = \{(A, V) : |A| < k\}$ , for which  $\mathcal{T}^+ = \emptyset$ .

Since all singletons  $\{(B, A)\}$  in  $\mathcal{T}^*$  satisfy  $B = V$ , by definition of  $\mathcal{T}$ , our sets  $O$  contain the corresponding separations  $(A, B)$ . So the relationship between each  $O$  and the  $\mathcal{T}^+$  it gives rise to is precisely

$$O = \{(A, B) \in \overrightarrow{S}_k : \{(B, A)\} \in \mathcal{T}^* \cup \mathcal{T}^+\}. \quad (*)$$

If  $O$  contains a separation  $(X, Y)$  together with its inverse  $(Y, X)$ , then  $(T, \alpha)$  with  $T = K^2$  and  $\alpha: \vec{E}(T) \rightarrow \{(X, Y), (Y, X)\}$  satisfies (2), by (\*). We now assume that  $O$  is antisymmetric: that it contains no inverse pair of separations. Then  $O$  is a consistent orientation of some subset  $S_O$  of  $S_k$ .

Let us prove by induction on  $|S_k \setminus S_O|$  that  $\mathcal{T}^+$  satisfies (1) or (2). At the induction start,  $O$  is an orientation of all of  $S_k$ . Hence if (1) fails then  $O$  has a subset  $\sigma \in \mathcal{T}^* \cup \mathcal{T}^+$ . By (\*) and the antisymmetry of  $O$ , we have  $|\sigma| \geq 2$ . Let  $T$  be the star  $K_{1,n}$  with  $n = |\sigma|$  leaves, and let  $\alpha$  map its oriented edges  $(e, s, t)$  with  $s$  a leaf bijectively to the elements of  $\sigma$ . Then  $(T, \alpha)$  satisfies (2), by definition of  $\sigma$  and (\*).

In the induction step we have  $S_k \setminus S_O \neq \emptyset$ . Choose  $\{U_1, W_1\}$  and  $\{U_2, W_2\}$  in  $S_k \setminus S_O$  so that both  $(U_i, W_i)$  are maximal in  $\overrightarrow{S}_k \setminus \overrightarrow{S}_O$  and  $(U_1, W_1) \geq (W_2, U_2)$ .<sup>10</sup> Then the  $(U_i, W_i)$  are maximal even in  $\overrightarrow{S}_k \setminus O$ :

<sup>10</sup> It is easy to see that such separations exist, just choose them in turn.

for any  $(U, W) > (U_i, W_i)$  in  $\vec{S}_k \setminus O$  we would have  $(W, U) \in O$  by the maximality of  $(U_i, W_i)$  in  $\vec{S}_k \setminus \vec{S}_O$ , so  $(W_i, U_i) > (W, U)$  would be in  $O$  (this being closed upwards in  $\vec{S}_k$ ), contradicting the fact that  $(U_i, W_i) \notin \vec{S}_O$ .

Thus, the sets  $O_i := O \cup \{(U_i, W_i)\}$  are again closed upwards in  $\vec{S}_k$ , and are orientations of subsets of  $S_k$  that contain  $S_O$  properly. We may therefore apply the induction hypothesis to these  $O_i$ , to obtain (1) or (2) for  $\mathcal{T}_i^+ := \{\{(B, A)\} : (A, B) \in O_i\} \setminus \mathcal{T}^* \supseteq \mathcal{T}^+$  instead of  $\mathcal{T}^+$ .

Since (1) holds with  $\mathcal{T}^+$  as soon as it holds with  $\mathcal{T}_1^+$  or  $\mathcal{T}_2^+$ , we may assume that both  $\mathcal{T}_i^+$  satisfy (2). Let  $(T_i, \alpha_i)$  be the corresponding  $S_k$ -trees over  $\mathcal{T}^* \cup \mathcal{T}_i^+$ . If one of these is in fact over  $\mathcal{T}^* \cup \mathcal{T}^+$  we are done, so we assume not. Then each  $T_i$  has a node  $u_i$  such that  $\alpha_i(e_i, w_i, u_i) = (W_i, U_i)$  for every edge  $e_i = w_i u_i$  of  $T_i$  at  $u_i$ .

Every such  $u_i$  must be a leaf. For otherwise  $(W_i, U_i) \geq (U_i, W_i)$ , and therefore  $W_i \subseteq U_i$ . But then  $U_i = V$  and hence  $(W_i, U_i) \in O$ , contrary to the choice of  $(U_i, W_i)$ . Similarly, these leaves  $u_i$  are unique. Indeed if  $u'_i$  is another leaf, with incident edge  $e'_i = u'_i w'_i$  say, then  $(e'_i, u'_i, w'_i) \geq (e_i, w_i, u_i)$  and therefore  $\alpha_i(e'_i, u'_i, w'_i) \geq \alpha_i(e_i, w_i, u_i) = (W_i, U_i)$ . Hence if  $\alpha_i(e'_i, u'_i, w'_i) = (U_i, W_i)$  then  $(U_i, W_i) \geq (W_i, U_i)$ , with a contradiction as earlier. Thus, our  $S_k$ -trees  $(T_i, \alpha_i)$  are nearly over  $\mathcal{T}^* \cup \mathcal{T}^+$ : they are, except at their leaf  $u_i$ .

Choose  $\{X_1, X_2\} \in S_k$  of minimum order with

$$(U_1, W_1) \geq (X_1, X_2) \geq (W_2, U_2);$$

such a separation exists, because  $(U_1, W_1)$  is a candidate. We shall modify the maps  $\alpha_i$  to maps  $\alpha'_i$  defining  $S_k$ -trees  $(T_i, \alpha'_i)$ . These will again be over  $\mathcal{T}^* \cup \mathcal{T}^+$  except at  $u_i$ , where we shall have  $\alpha'_i(e_i, w_i, u_i) = (X_{3-i}, X_i)$ . Our plan will then be to join the newly labelled trees  $T_i - u_i$  together by adding the edge  $w_i w_{3-i}$  and mapping it to their common separation  $(X_{3-i}, X_i)$ , to obtain our desired  $S_k$ -tree over  $\mathcal{T}^* \cup \mathcal{T}^+$ .

To define  $\alpha'_i$ , consider an edge  $e$  of  $T_i$ . Name its ends  $t, t'$  so that  $(e_i, u_i, w_i) \geq (e, t, t')$ . Then if  $\alpha_i(e, t, t') = (A, B)$ , say, let

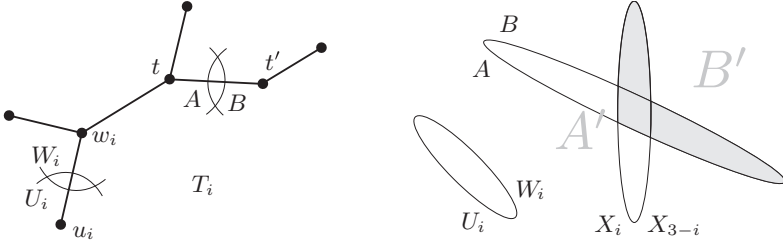
$$\alpha'_i(e, t, t') = (A', B') := (A \cup X_i, B \cap X_{3-i})$$

and  $\alpha'_i(e, t', t) = (B', A')$ . (Fig. 12.5.6).

Let us show that  $\alpha'_i$  maps  $\vec{E}(T_i)$  to  $\vec{S}_k$ . Given an edge  $e \in E(T_i)$ , let  $\vec{e}$  be such that  $(e_i, u_i, w_i) \geq \vec{e}$ , as in the definition of  $\alpha'_i$ . Then for  $\alpha_i(\vec{e}) = (A, B)$  we have  $\alpha'_i(\vec{e}) = (A', B') = (A \cup X_i, B \cap X_{3-i})$ . By  $(\dagger)$  we have  $|A' \cap B'| \leq |A \cap B| < k$ , as desired, if the order of

$$(Y_i, Y_{3-i}) := (A \cap X_i, B \cup X_{3-i})$$

is no less than the order of  $\{X_1, X_2\}$ . And it cannot be less, as that would contradict our choice of  $\{X_1, X_2\}$  since  $(U_1, W_1) \geq (Y_1, Y_2) \geq (W_2, U_2)$ .

Fig. 12.5.6. Shifting  $(A, B)$  to  $(A', B')$ 

Indeed, recall that  $U_i \subseteq X_i$  as well as  $U_i \subseteq A$ , and  $W_i \supseteq X_{3-i}$  as well as  $W_i \supseteq B$ , because  $(U_i, W_i) \geq (X_i, X_{3-i})$  as well as  $(U_i, W_i) \geq (A, B)$ . Hence  $(U_i, W_i) \geq (Y_i, Y_{3-i})$ , and  $(Y_i, Y_{3-i}) \geq (X_i, X_{3-i}) \geq (W_{3-i}, U_{3-i})$ .

Next, let us show that  $\alpha'_i$ , like  $\alpha_i$ , respects the orderings of  $\vec{E}(T_i)$  and  $\vec{S}_k$ . Consider any  $\vec{e} \geq \vec{f} \in \vec{E}(T_i)$ , with  $(A, B) = \alpha_i(\vec{e}) \geq \alpha_i(\vec{f}) = (C, D)$  say. Then  $A \subseteq C$  and  $B \supseteq D$ . If  $(e_i, u_i, w_i) \geq \vec{e}$ , then  $\alpha'_i(\vec{e}) =: (A', B') \geq (C', D') := \alpha'_i(\vec{f})$ , because  $A' = A \cup X_i \subseteq C \cup X_i = C'$  and  $B' = B \cap X_{3-i} \supseteq D \cap X_{3-i} = D'$ . On the other hand if  $(e_i, u_i, w_i) \geq \vec{e}$  but  $(e_i, u_i, w_i) \geq \vec{f}$ , then  $A' = A \cap X_{3-i} \subseteq A \subseteq C \subseteq C \cup X_i = C'$  while  $B' = B \cup X_i \supseteq B \supseteq D \supseteq D \cap X_{3-i} = D'$ , so again  $(A', B') \geq (C', D')$ . Since  $\alpha'_i$ , by definition, commutes with inversions of orientations, this covers all the cases to be considered.

So  $(T_i, \alpha'_i)$  is indeed an  $S_k$ -tree. Let us show that  $(T_i, \alpha'_i)$ , like  $(T_i, \alpha_i)$ , is an  $S_k$ -tree over  $\mathcal{T}^* \cup \mathcal{T}^+$  except at  $u_i$ , where

$$\alpha'_i(e_i, w_i, u_i) = (W_i \cap X_{3-i}, U_i \cup X_i) = (X_{3-i}, X_i).$$

Consider a node  $t \neq u_i$ , and let  $e = st$  be its incident edge in  $u_i T_i t$ . Let  $\alpha_i(e, s, t) =: (A, B)$  and  $\alpha'_i(e, s, t) =: (A', B')$ . As  $(e_i, u_i, w_i) \geq (e, s, t)$ ,

$$(U_i, W_i) \geq (A, B) \geq (A \cup X_i, B \cap X_{3-i}) = (A', B').$$

Suppose first that  $\alpha_i(\sigma_t) = \{(A, B)\}$ . Let us show that  $t$  must be a leaf.

If not, then  $\alpha_i(e', s', t) = (A, B)$  for another edge  $e' = s't$  at  $t$ . Then  $(A, B) \geq (B, A)$ , and hence  $B = V$ . But then  $(A, B) \in O$ , and hence  $(U_i, W_i) \in O$  since  $O$  is closed upwards in  $\vec{S}_k$ . This contradicts our choice of  $(U_i, W_i)$ . Hence  $t$  is indeed a leaf of  $T_i$ .

Since  $(T_i, \alpha_i)$  is over  $\mathcal{T}^* \cup \mathcal{T}^+$  except at  $u_i$ , we thus have  $(B, A) \in O$  by (\*). As  $O$  is closed upwards in  $\vec{S}_k$ , this means that  $(B', A') \geq (B, A)$  is also in  $O$ . Thus,  $\alpha'_i(\sigma_t) = \{(A', B')\} \in \mathcal{T}^* \cup \mathcal{T}^+$  by (\*), as desired.

Suppose now that  $|\alpha_i(\sigma_t)| \geq 2$ . Then  $\alpha_i(\sigma_t) \in \mathcal{T}^*$ ; let us show that also  $\alpha'_i(\sigma_t) \in \mathcal{T}^*$ . Since  $\alpha'_i$  respects the orderings of  $\vec{E}(T_i)$  and  $\vec{S}_k$ , we already know that  $\alpha'_i(\sigma_t)$  is a star. To show that  $\alpha'_i(\sigma_t) \in \mathcal{T}$ , consider any edge  $e' = s't \neq st$ . Let  $(C, D) := \alpha_i(e', s', t)$  and  $(C', D') := \alpha'_i(e', s', t)$ .

Then  $(e_i, u_i, w_i) \geq (e', t, s')$  by the choice of  $e \neq e'$ . Hence  $C' = C \cap X_{3-i}$ , but also  $A' = A \cup X_i$ . Thus, any vertex or edge of  $G[C']$  that is not in  $G[C']$  lies in  $G[X_i] \subseteq G[A']$ . The fact that  $\alpha_i(\sigma_t) \in \mathcal{T}$  thus implies  $\alpha'_i(\sigma_t) \in \mathcal{T}$ , as desired.

Let  $T$  be the tree obtained from the disjoint union of  $T_1 - u_1$  and  $T_2 - u_2$  by joining  $w_1$  to  $w_2$  by a new edge  $e$ . Let  $\alpha: \vec{E}(T) \rightarrow \vec{S}_k$  map  $(e, w_{3-i}, w_i)$  to  $\alpha'_i(e_i, u_i, w_i) = (X_i, X_{3-i})$  for  $i = 1, 2$ , and otherwise extend the  $\alpha'_i$ . Then  $\alpha$  commutes with the inversion of  $\vec{e}$  and of  $(X_1, X_2)$ , and  $\alpha(\sigma_t) = \alpha'_i(\sigma_t) \in \mathcal{T}^* \cup \mathcal{T}^+$  for all  $t \in T$ , in particular for  $t = w_i$ . Hence,  $(T, \alpha)$  satisfies (2).  $\square$

Theorem 12.5.9, as stated above, is a special case of a more general result in which  $\mathcal{T}^*$  can be replaced by other collections  $\mathcal{F}$  of ‘forbidden’ stars of oriented separations. Given a set  $S$  of separations, an  $\mathcal{F}$ -tangle of  $S$  is a consistent orientation of  $S$  that has no subset  $\sigma \in \mathcal{F}$ . For

 $\mathcal{F}$ -tangle

$$\mathcal{F}_k := \left\{ \sigma \subseteq \vec{S} : \sigma \text{ is a star and } \left| \bigcap \{ B : (A, B) \in \sigma \} \right| < k \right\}$$

we obtain the following more tangle-like duality theorem for tree-width:

**Theorem 12.5.11.** *The following two assertions are equivalent for all graphs  $G$  and integers  $k > 0$ :*

- (i)  $G$  has tree-width less than  $k - 1$ ;
- (ii)  $G$  has no  $\mathcal{F}_k$ -tangle of  $S_k$ .

It is not hard to show that (i) is equivalent to the existence of an  $S_k$ -tree over  $\mathcal{F}_k$  in  $G$  (Ex. 59). Theorem 12.5.11 in that form and Theorem 12.5.9 are both corollaries of a more general theorem about tangle-tree duality in so-called ‘abstract separation systems’; see the notes.

Moreover, any  $\mathcal{F}_k$ -tangle of  $S_k$  gives rise to a  $k$ -bramble (Ex. 59). Thus, Theorem 12.5.11 implies the tree-width duality theorem (12.4.3).

## 12.6 Tree-decompositions and forbidden minors

If  $\mathcal{H}$  is any set or class of graphs, then the class

$$\text{Forb}_{\preceq}(\mathcal{H}) := \{ G \mid G \not\preceq H \text{ for all } H \in \mathcal{H} \}$$

 $\text{Forb}_{\preceq}(\mathcal{H})$ 

of all graphs without a minor in  $\mathcal{H}$  is a graph property, i.e. is closed under isomorphism.<sup>11</sup> When it is written as above, we say that this property is expressed by specifying the graphs  $H \in \mathcal{H}$  as *forbidden* (or *excluded*) *minors*.

forbidden  
minors

<sup>11</sup> As usual, we abbreviate  $\text{Forb}_{\preceq}(\{H\})$  to  $\text{Forb}_{\preceq}(H)$ .